Fiscal Year:	FY 2021	**Task Last Updated:**	FY 07/22/2021
PI Name:	Jinkerson, Robert Ph.D.		
Project Title:	Genetically Minimizing Non-Edible Portions of Plants for Space Flight Applications		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92507	**Congressional District:**	41
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	01/01/2019	**End Date:**	12/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Orozco-Cardenas, Martha Ph.D. (University of California, Riverside)		
Grant/Contract No.:	NNX16AO69A-T0410		
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Performance Goal Text:			

Task Description:	Plants on space expeditions can provide a fresh source of food and nutrients, carbon dioxide uptake capacity, and behavioral health benefits to crew members. Most fruit and vegetable bearing plants are too large and produce too much inedible biomass for space flight applications. This has limited plants grown in space to leafy green vegetables, such as lettuce. Increasing the diversity of plants suitable for space flight would increase nutritional opportunities and increase the diversity of fresh food options available for crew members. We have developed a genetically engineered tomato plant that is altered in its developmental cycle that is ideal for growth in space flight applications. These plants grow minimal amounts of leaves, flower, and develop fruit in rapid progression. These tomato plants have several traits that make them ideal for cultivation on a spacecraft: 1) small size, 2) very small amount of non-edible biomass produced, and 3) the ability to produce fruit faster. This project aims to (1) develop this trait, which we are calling Small Plants for Agriculture in Confined Environments (SPACE), in tomatoes and determine the feasibility of these plants to be grown on the International Space Station (ISS) in NASA's Vegetable Production System (Veggie), (2) engineer key components of an algal carbon concentrating mechanism into SPACE tomato plants to increase carbon capture and growth rates, and (3) evaluate nutrient levels, palatability, and microbial loading levels of SPACE tomato fruit grown in simulated ISS conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Plants on extended space expeditions can provide a fresh source of food and nutrients, CO2 uptake capacity, and behavioral health benefits to crew members. Most fruit and vegetable bearing plants, however, are too large and produce too much inedible biomass for space flight applications. For this project, we are evaluating a mutant tomato that is an extreme dwarf and can revolutionize the way crops are grown in space. Plants bearing this phenotype are referred to as Small Plants for Agriculture in Confined Environments (SPACE) plants. The impact these SPACE tomatoes could have for space flight applications include: 1) SPACE tomatoes could provide a ready to eat fruit option for astronauts on extend space missions, 2) SPACE tomatoes have reduced non-edible biomass compared to other tomato varieties, and 3) Similar mutations could be generated in other crops (peppers, eggplant, carrot) to decrease plant size. On Earth these SPACE tomatoes: 1) can revolutionize indoor / vertical agriculture, 2) are generating interest in commercial licensing from several companies, and 3) can accelerate plant research.
Task Progress:	Plants on space expeditions can provide a fresh source of food and nutrients, carbon dioxide uptake capacity, and behavioral health benefits to crew members. Most fruit and vegetable bearing plants are too large and produce too much inedible biomass for space flight applications. This has limited plants grown in space to leafy green vegetables, such as lettuce. Increasing the diversity of plants suitable for space flight would increase nutritional opportunities and increase the diversity of fresh food options available for crew members. We have developed a genetically engineered tomato plant that is altered in its developmental cycle that is ideal for growth in space flight applications. These plants grow minimal amounts of leaves, flower, and develop fruit in rapid progression. These tomato plants have several traits that make them ideal for cultivation on a spacecraft: (1) small size, (2) very small amount of non-edible biomass produced, and (3) the ability to produce fruit faster. This project aimed to (1) develop this trait, which we are calling Small Plants for Agriculture in Confined Environments (SPACE), in tomatoes and determine the feasibility of these plants to be grown on the International Space Station (ISS) in NASA's Vegetable Production System (Veggie), (2) engineer key components of an algal carbon concentrating mechanism into SPACE tomato plants to increase carbon capture and growth rates, and (3) evaluate nutrient levels, palatability, and microbial loading levels of SPACE tomato fruit grown in simulated ISS conditions. When we first started the project, it was unclear if the SPACE phenotype was a result of transformation or can be reproduced anytime. During the project we demonstrated that SPACE tomatoes can be cultivated reproducibly both in vitro and in soil. The SPACE phenotype can manifest from seed germination or from tissue culture regeneration. Mutant plants were genotyped and two mutant genotypes were identified that lead to the SPACE phenotype. We also isolated homozygous lines from the T1 and T2 plants. A qPCR protocol was established to facilitate rapid genotyping of plants. Mutant plants were grown and their phenotypes determined. Harvest index, a plant productivity metric used to describe the relative distribution of biomass between the edible and inedible components of a crop (Hay, 1995), was determined for both wildtype and SPACE tomato plants. The SPACE tomato plants have an almost 30% increase in harvest index. We measured other physiological metrics such as time to fruit, flowering history, flower bud history, fruiting history, height, leaf number, and senescent leaves history. From all of this data we performed two rounds of selection of the inbred lines to select those with the most desirable traits. Currently we are at the T4 generation. From these selected lines, plant growth was scaled up to allow us to grow a large number of plants to generate seed stock and evaluate the fruit. We performed tests to evaluate the SPACE tomato fruit for nutrient levels and palatability. No difference was found between SPACE tomato fruit and wildtype tomato fruit, suggesting that decreases in plant size do not alter fruit taste or nutrient levels. To increase CO2 fixation in the SPACE plants we have prepared plasmids with genes from algae that are key for the algal carbon concentrating mechanism. These algal genes have been transformed into plants but could not be evaluated before the end of the grant due to delays resulting from COVID-19. Overall these findings have helped to move SPACE tomatoes towards being a viable plant for cultivation during space flight. The future for the SPACE tomato project is promising. In the last year, our proposal to grow the SPACE tomatoes on the ISS in in the Advanced Plant Habitat was selected by NASA Space Biology. This grant, starting March 1, 2021, aims to demonstrate a seed-to-seed growth experiment on the ISS, an important step towards crop systems in space. Here on Earth the SPACE tomatoes will be evaluated for use in vertical controlled environment agriculture. We received a three-year grant from the Foundation for Food and Agriculture Research (FFAR) to evaluate SPACE tomatoes in vertical farms. Additionally, several vertical agriculture companies have expressed interest in licensing the SPACE tomatoes and in forming collaborations with us to conduct growth trials of the plants in their vertical farms. Reference: Hay RKM. Harvest index: a review of its use in plant breeding and crop physiology. Annals of Applied Biology. 1995 Feb;126:197-216.
Bibliography Type:	Description: (Last Updated: 07/05/2023)

Articles in Peer-reviewed Journals	Hann EC, Overa S, Harland-Dunaway M, Narvaez AF, Le DN, Orozco-Cárdenas ML, Jiao F, Jinkerson RE. "A hybrid inorganic–biological artificial photosynthesis system for energy-efficient food production." Nature Food. 2022 Jun 23;3:461-71. https://doi.org/10.1038/s43016-022-00530-x ; PMID: 37118051 , Jun-2022